Simulation Output Summary

Integrated Energy Systems Optimiser (IESO) v24.03 · www.github.com/greoux-research/ieso

March 25, 2024 09:18 AM

Demand

Demand for Electricity:

- Annual Demand in MWh: 50000000
- Hourly Profile: Figure 1
- Non-Service Penalty in $ per MWh: 20000
- Supply Sources: Power Grid

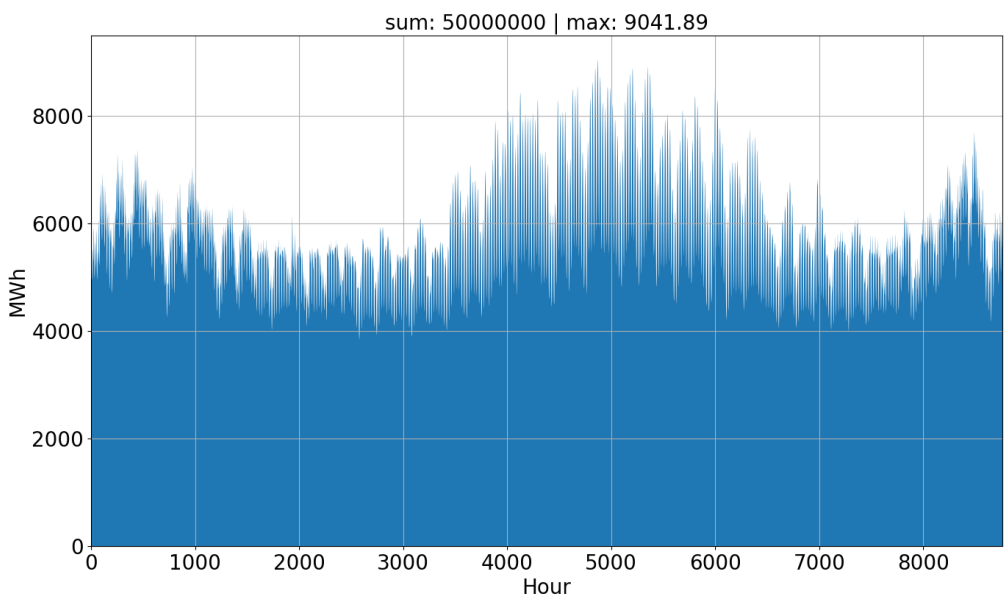

Figure 1: Demand for Electricity

Demand for Water:

- Annual Demand in Q(X): 365000000

- Hourly Profile: Uniform (41667 Q(X) each Hour)
- Non-Service Penalty in $ per Q(X): 1000
- Supply Sources: Power-to-X Processes ["p2x-med"]

Generators

Generator "nucl":

- Type: Cogenerator ($a = 0.12$; $b = 1.98$)
- Fixed Costs in $ per MW per Year: 456466
- Variable Costs in $ per MWh: 11.5
- Variable Emissions in $ per MWh: 0
- The optimiser did not suggest adding generation capacity

Generator "coal":

- Type: Cogenerator ($a = 0.12$; $b = 1.57$)
- Fixed Costs in $ per MW per Year: 222392
- Variable Costs in $ per MWh: 26.84
- Variable Emissions in $ per MWh: 880
- Capacity in MW: 2587.75
- Electricity Generation: 13378007 MWh (Figure 2)
- Heat Generation: 13792916 MWh (Figure 3)
- Number of Successive Changes in Output: 4055

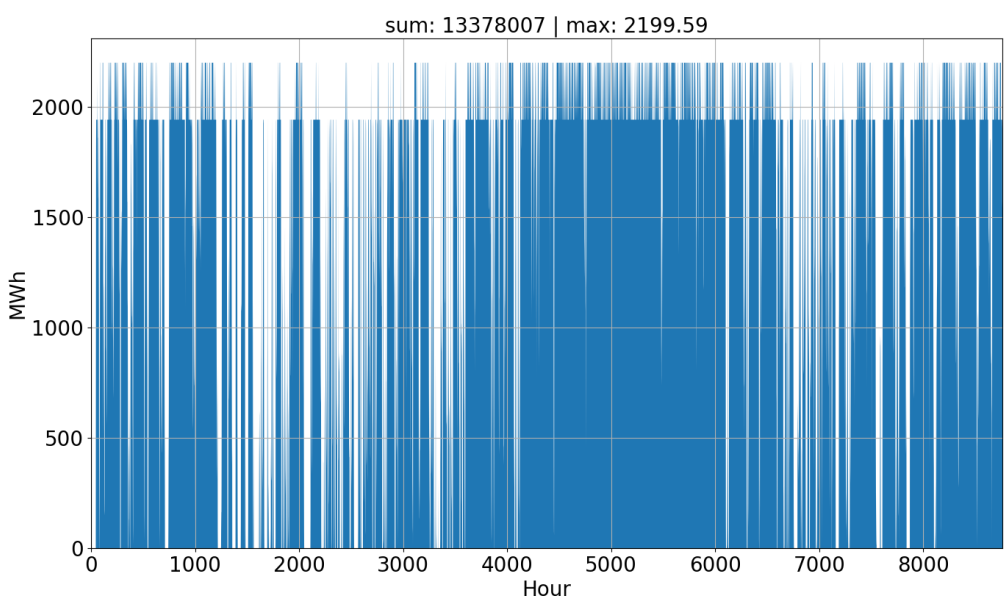

Figure 2: Electricity Generation by "coal"

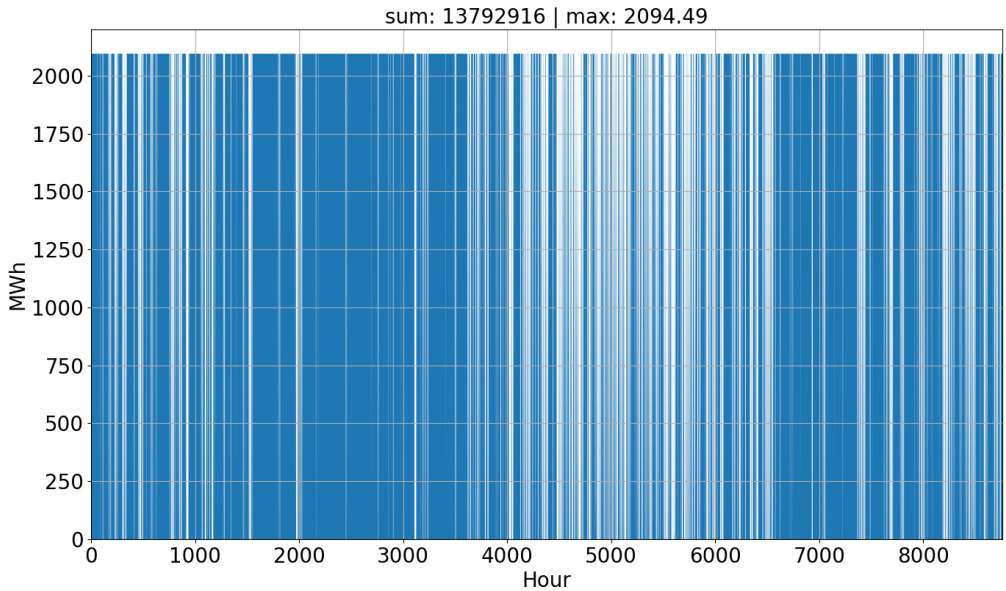

Figure 3: Heat Generation by "coal"

Generator "ccgt":

- Type: Cogenerator ($a = 0.12$; $b = 1.57$)
- Fixed Costs in $ per MW per Year: 98749
- Variable Costs in $ per MWh: 56.44
- Variable Emissions in $ per MWh: 365
- Capacity in MW: 2950.34
- Electricity Generation: 6032243 MWh (Figure 4)
- Heat Generation: 4457084 MWh (Figure 5)
- Number of Successive Changes in Output: 4190

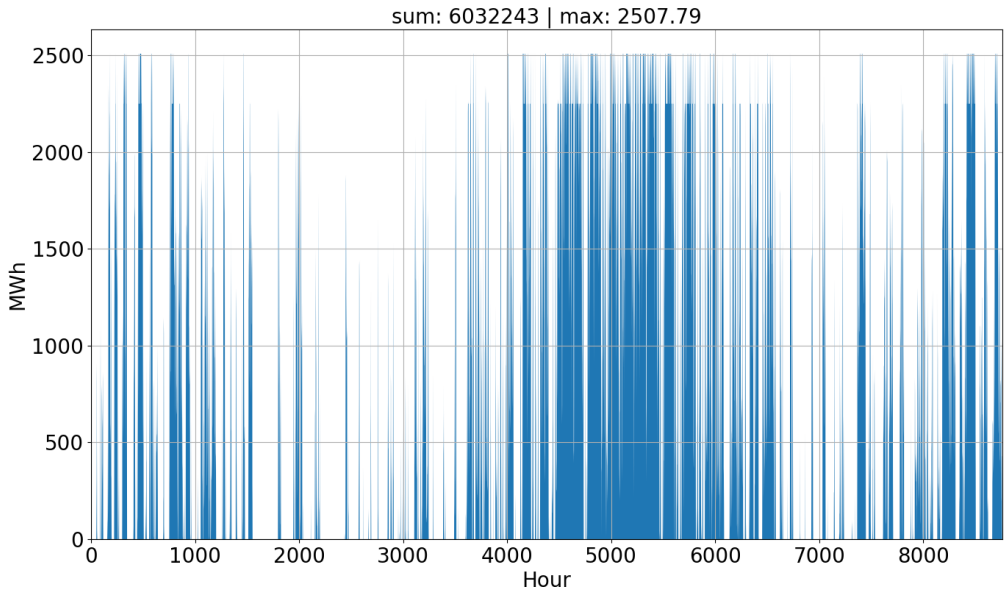

Figure 4: Electricity Generation by "ccgt"

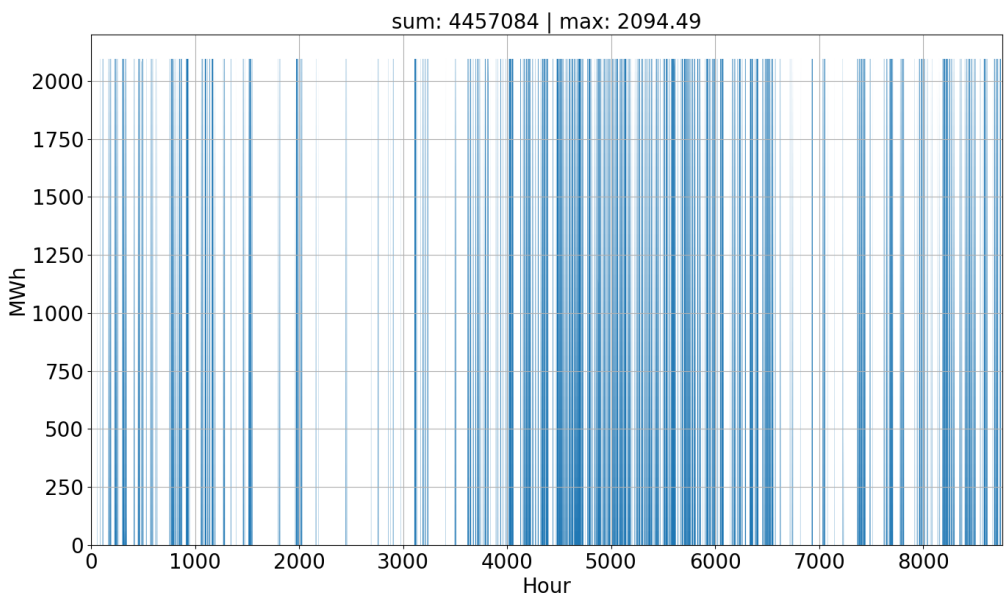

Figure 5: Heat Generation by "ccgt"

Generator "ocgt":

- Type: Electricity-Generating Power Plant

- Fixed Costs in $ per MW per Year: 75182
- Variable Costs in $ per MWh: 96.11
- Variable Emissions in $ per MWh: 523
- Capacity in MW: 1902.71
- Electricity Generation: 364604 MWh (Figure 6)
- Number of Successive Changes in Output: 592

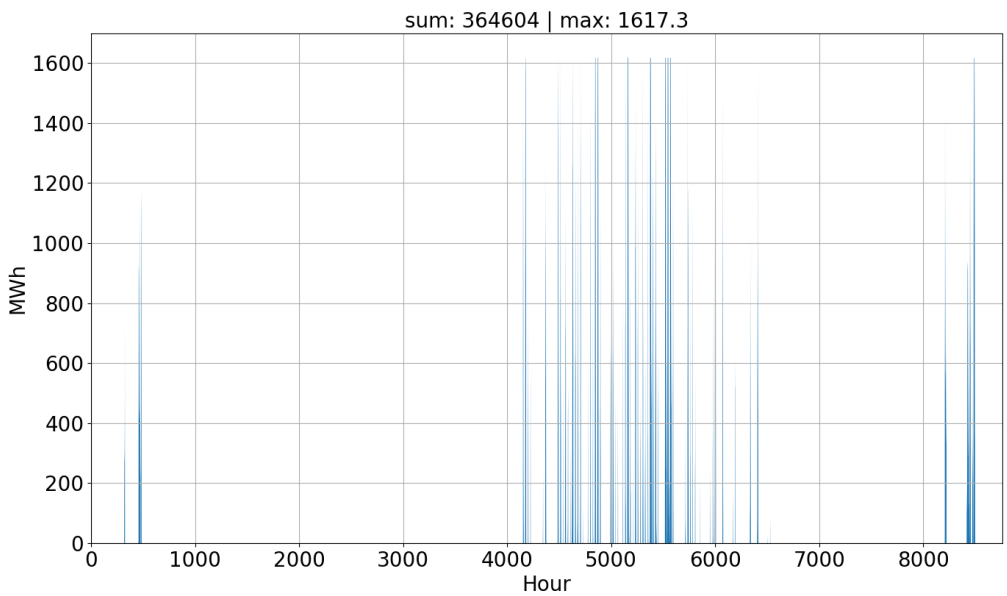

Figure 6: Electricity Generation by "ocgt"

Generator "solr":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 70046
- Variable Costs in $ per MWh: 3e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 5032.63
- Electricity Generation: 8273577 MWh (Figure 7)
- Curtailment: 0 Hours; 0 MWh or 0.0 % of the Tech.'s Annual Gen. Volume

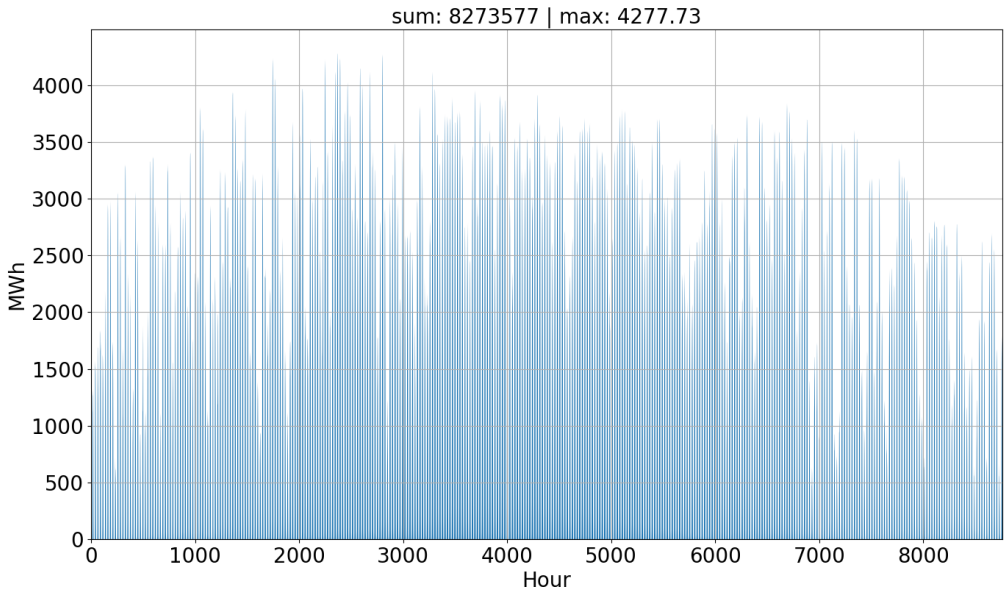

Figure 7: Electricity Generation by "solr"

Generator "wind":

- Type: Electricity-Generating Power Plant
- Fixed Costs in $ per MW per Year: 133553
- Variable Costs in $ per MWh: 6e-06
- Variable Emissions in $ per MWh: 0
- Capacity in MW: 6742.49
- Electricity Generation: 22628370 MWh (Figure 8)
- Curtailment: 595 Hours; 550347.01 MWh or 2.43 % of the Tech.'s Annual Gen. Volume

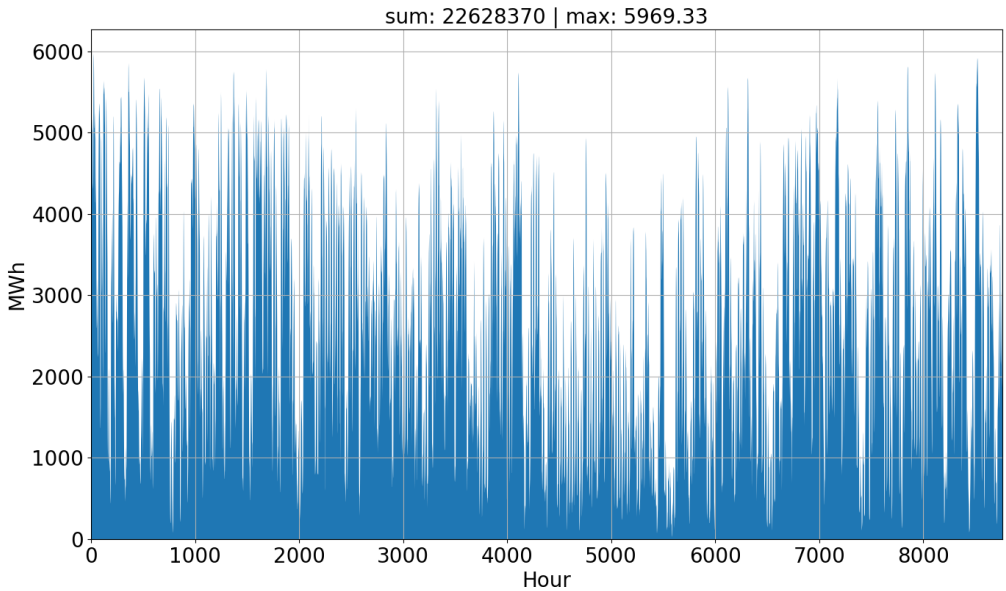

Figure 8: Electricity Generation by "wind"

Flexibility Means

Flexibility Mean "bstr":

- Type: Electricity Storage System
- Fixed Costs in $ per MWh per Year: 27248
- Hours of Storage at Maximum Discharge: 4
- Round Trip Efficiency: 0.85
- Capacity in MWh: 3648.53
- Electricity Charged (-) and Discharged (+): Figure 9
- Electricity Stored: Figure 10

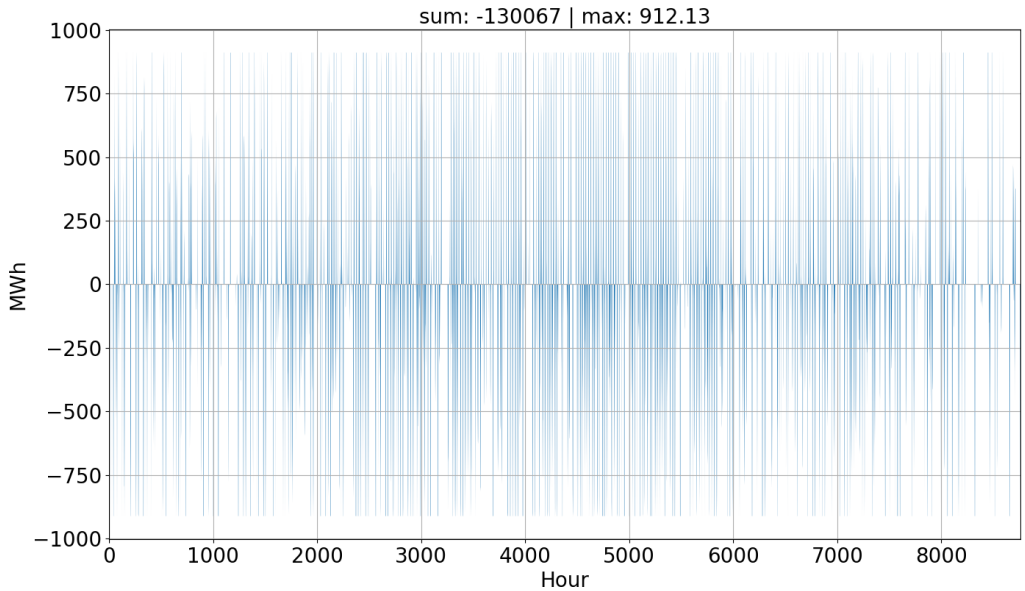

Figure 9: Electricity Charged (-) and Discharged (+) by "bstr"

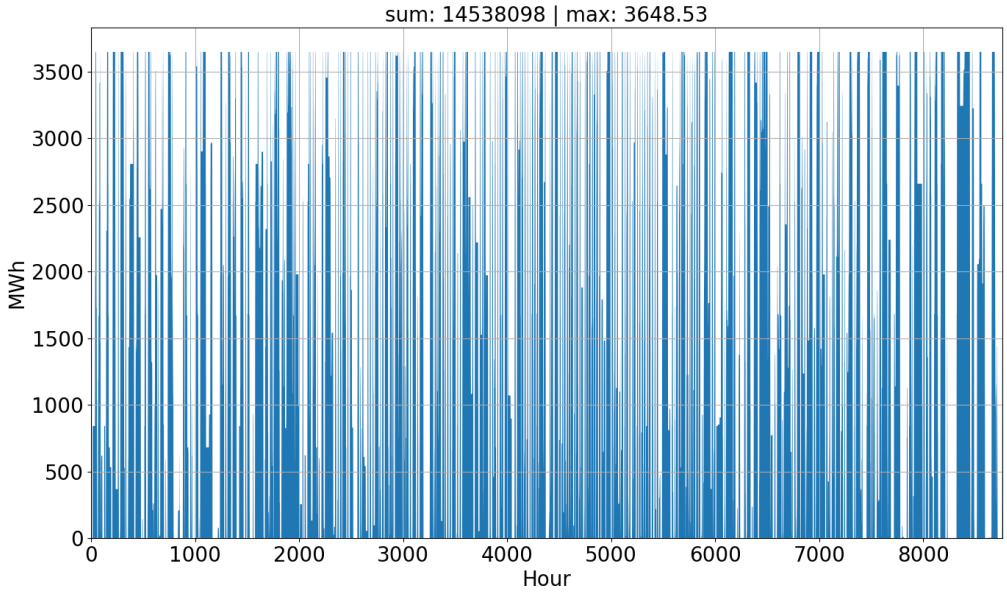

Figure 10: Electricity Stored by "bstr"

Power-to-X Processes

Power-to-X Process "p2x-med":

- Consumes Electricity (Grid-Supplied 0.0015 MWh per Q(X))
- Consumes Heat (0.05 MWh per Q(X) at 80 C)
- Heat Supply Sources: Cogenerators ["nucl" , "coal" , "ccgt"]
- Fixed Production Costs in $ per Q(X) per Hour per Year: 2185
- Variable Production Costs (Excluding Energy) in $ per Q(X): 0.26
- Production Capacity in Q(X) per Hour: 49282.02
- Production Profile: Figure 11
- Fixed Storage Costs in $ per Q(X) per Year: 0.22
- Storage Capacity: 705819.63 Q(X) or 17 Hours of Production at Max. Capacity
- Storage Profile: Figure 12
- Supply Profile: Figure 13

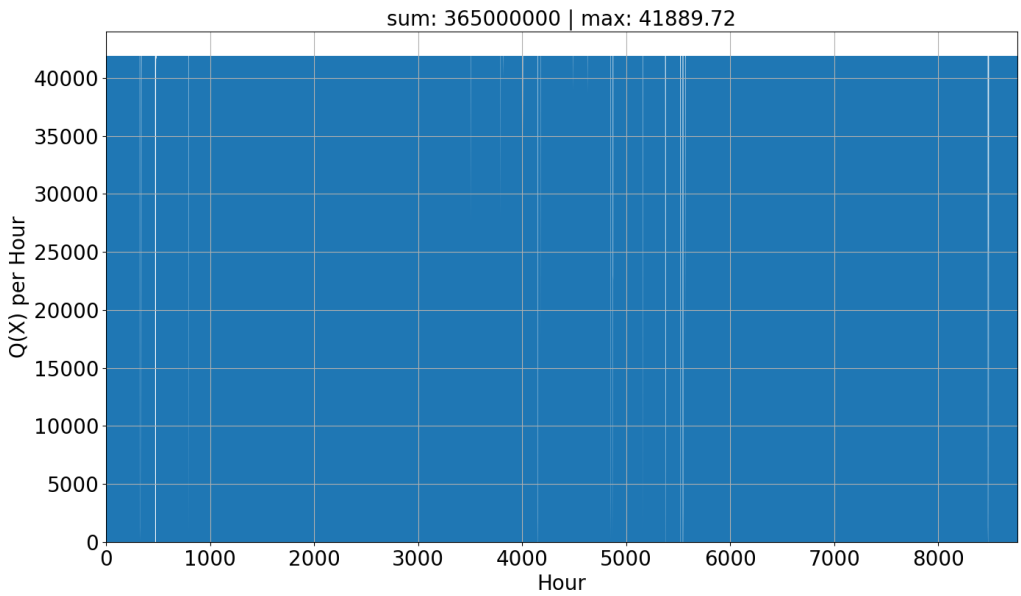

Figure 11: Production Profile - "p2x-med"

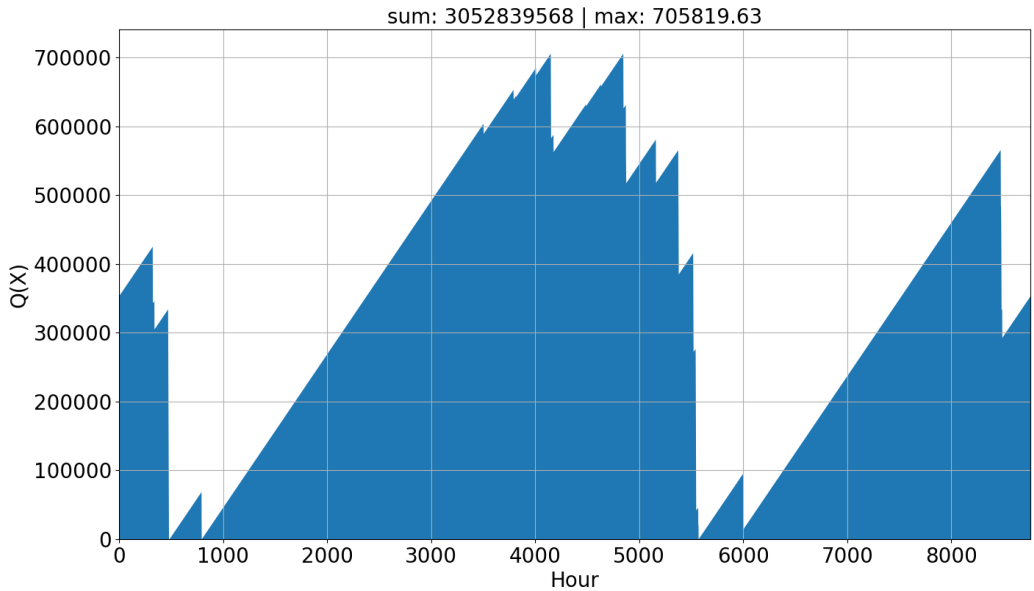

Figure 12: Storage Profile - "p2x-med"

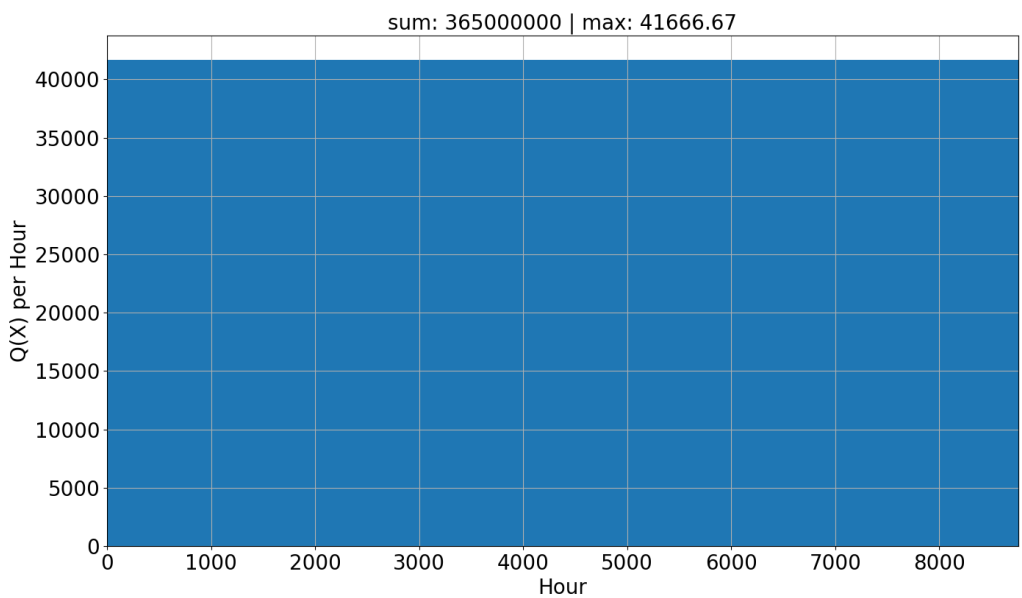

Figure 13: Supply Profile - "p2x-med"

Costs and Emissions

Generation Costs and Emissions on a per MWh Basis:

- Generation Costs in $ per MWh: 58.08
- Generation Emissions in kg per MWh: 299.7

Prices

Electricity:

- Average Price in $ per MWh: 61.21
- Price Profile: Figure 14

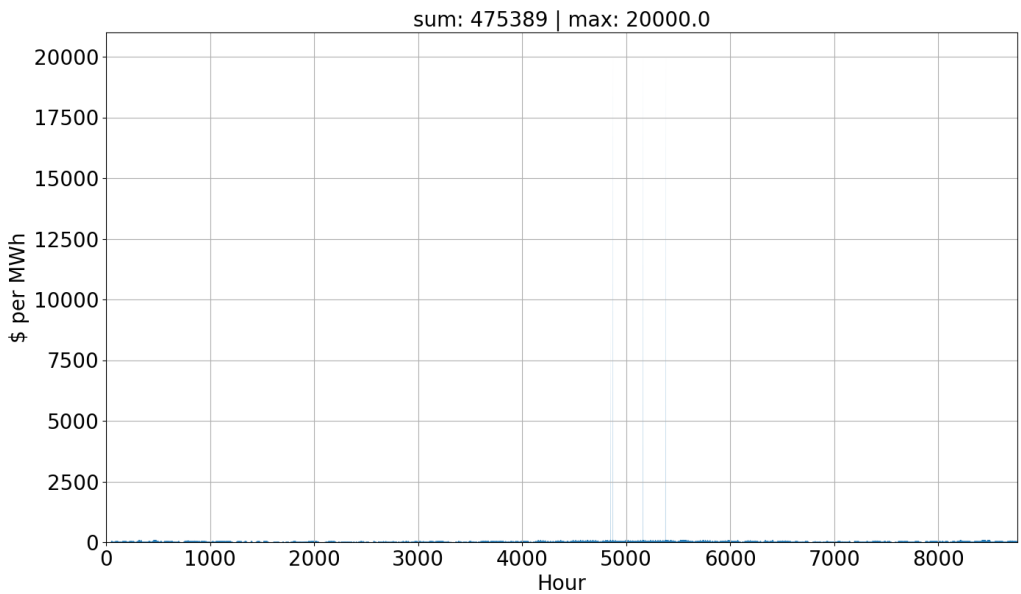

Figure 14: Electricity Price Profile

Water:

- Average Price in $ per Q(X): 0.91
- Price Profile: Figure 15

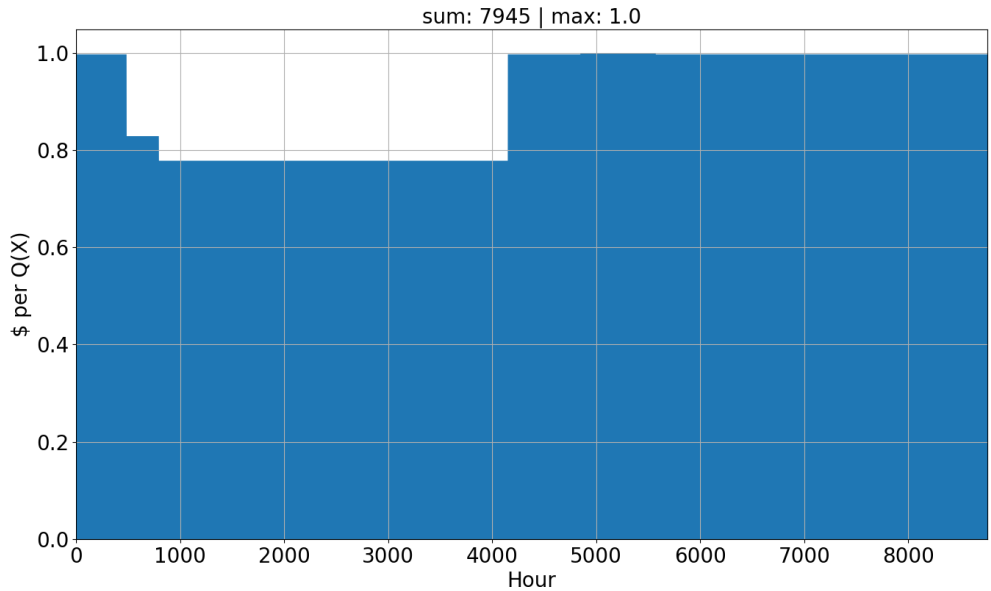

Figure 15: Water Price Profile

"p2x-med" Heat Supply:

- Average Price in $ per MWh: 5.71
- Price Profile: Figure 16

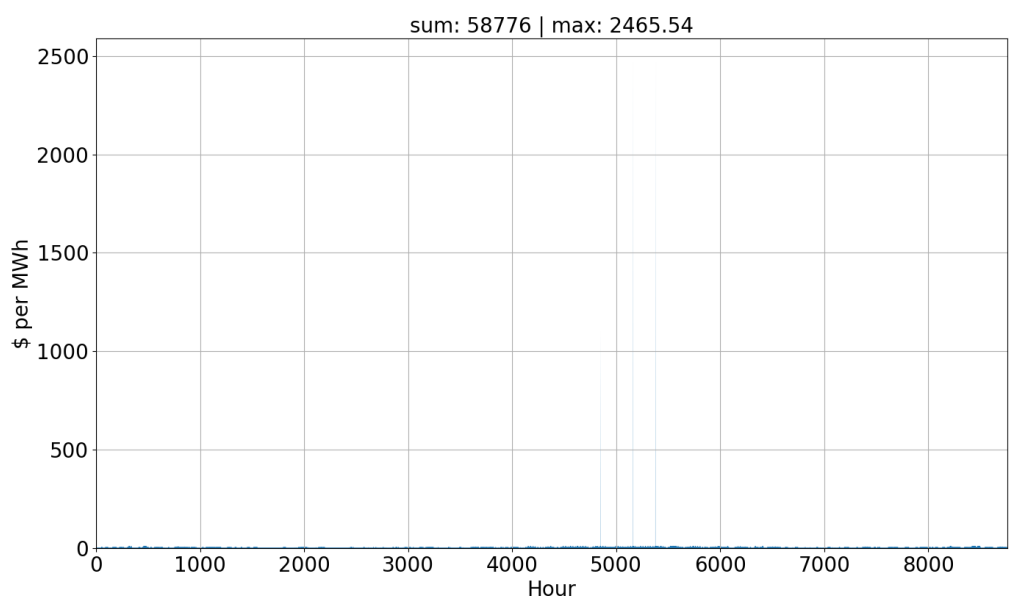

Figure 16: "p2x-med" Heat Supply Price Profile

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